TECHNICAL DILIGENCE REVIEW - RECORD 012

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-012

1. Scope of the safety review and the documents examined

Calder Rowe Engineering Advisory delivered report TDR-2025-41 on 2026-02-04; the scope agreed with Ardent Peak was the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet. As instructed, Calder Rowe Engineering Advisory confined this report to the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet; three adjacent questions raised during the work are listed as follow-ups rather than answered here. Two positions in TDR-2025-41 are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. The reviewer asked in each case whether a declined item had a compensating control, and TDR-2025-41 reports the answer even where it is unsatisfactory. Cross-references in this report point to the evidence index rather than to a summary of the evidence.

In the report at TDR-2025-41 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. K. Vandermeer was granted read access to the code, the pipeline and the incident tracker for the purposes of this report; production access was not requested. The present report distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. Where the report at TDR-2025-41 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. K. Vandermeer notes that the company answered every document request, though two answers arrived after the draft.

Interview evidence in the present report came from 15 sessions across 9 teams, each minuted and circulated for correction. K. Vandermeer interviewed 15 engineers and managers for report TDR-2025-41; two declined to be quoted on one topic and that is noted where it matters. Selection for the report at TDR-2025-41 favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. The present report added two components to its sample after the first week, because the interview evidence pointed at them. The company reviewed a draft of the report at TDR-2025-41 for factual accuracy and its corrections are reflected without altering any conclusion.

This report relies on the company's incident tracker for completeness of the incident population, which the reviewer could corroborate but not prove. The report at TDR-2025-41 does not include a code audit; the reviewer read code where a specific claim required it and no further. Work on the report at TDR-2025-41 ran over 7 weeks and included 15 interviews and read-only access to the engineering repositories. Method in this report was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. K. Vandermeer has stated the limitations at the point they bite rather than collecting them into a disclaimer.

The present report grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. Readers of this report should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. For Ardent Peak's purposes, the report at TDR-2025-41 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Drawing the work together, this report finds that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

12 services were examined in detail for report TDR-2025-41, chosen for their position in the dependency map rather than at random. Sampling in this report covered 12 of the 34 catalogued services and 7 of the 18 live sites. K. Vandermeer treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and report TDR-2025-41 keeps the two apart. The convention throughout the report at TDR-2025-41 is that an absence of evidence is reported as such rather than as evidence of absence. Calder Rowe Engineering Advisory has reported the position as at 2026-02-04 and has not attempted to predict where it will be at completion.

Schedule I - Interview programme and follow-up requests arising (TDR-2025-41)

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2026-07-10	Sam Oyelaran	head of engineering	Identity and Authorization	4	0
2026-05-27	Farida Cisse	head of engineering	Battery Management	2	1

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
			Service		
2026-04-18	Yuki Harada	engineering owner	Remote Operations Console	2	1
2026-09-28	Erik Palmgren	head of engineering	Lidar Driver	2	1
2025-12-06	Ines Duarte	release manager	Edge Gateway Agent	4	2
2026-05-03	Yuki Harada	security engineer	Remote Operations Console	4	4
2026-10-02	Nadia Semenova	site reliability lead	Fleet Allocator	3	2
2025-12-30	Sonia Krall	engineering owner	Lidar Driver	2	2
2026-05-08	Wanda Njoku	field reliability lead	Edge Gateway Agent	1	1
2026-09-02	Rahul Menon	site reliability lead	Release Orchestrator	4	4
2026-01-10	Rahul Menon	release manager	Localization Engine	4	3
2025-10-24	Rahul Menon	quality lead	OTA Updater	2	2
2026-06-15	Sam Oyelaran	safety engineer	Config Distribution	2	1
2026-05-28	Rahul Menon	release manager	Traffic Manager	5	3
2025-09-07	Ines Duarte	quality lead	Video Snippet Service	2	0
2026-02-19	Rahul Menon	head of engineering	Analytics Warehouse Loader	2	0
2026-04-23	Erik Palmgren	security engineer	Lidar Driver	1	1
2025-11-14	Sonia Krall	head of engineering	Fleet Allocator	4	1
2025-10-24	Wanda Njoku	security engineer	Battery Management Service	5	3
2025-09-22	Tomas Villareal	quality lead	Battery Management Service	5	5
2026-05-17	Marta Kellerman	safety engineer	Localization Engine	4	0
2025-07-29	Erik Palmgren	engineering owner	Charge Scheduler	4	1
2026-07-08	Bettina Vogel	quality lead	Gripper Controller	1	1
2026-05-28	Bettina Vogel	engineering owner	Robot Shadow Store	2	1
2026-02-07	Grace Lindqvist	quality lead	Motion Planner	4	4
2025-09-10	Owen Brantley	site reliability lead	Remote Operations Console	4	4
2026-06-09	Wanda Njoku	quality lead	Gripper Controller	2	2
2026-08-16	Colin Achebe	field reliability lead	Video Snippet Service	4	1
2026-09-24	Bettina Vogel	quality lead	Lidar Driver	4	4
2026-06-27	Sonia Krall	site reliability lead	Gripper Controller	3	3

2. Hazard identification: HZ-10 and the register as a whole

Hazard HZ-10 is the entry for the case that charging contact arcs during an unplanned disconnect. Status verified and residual risk Broadly acceptable are what the register gives for hazard HZ-10. Protection at Fire hazard HZ-10 comes from contactor sequencing with an arc-suppression design in the charge dock. HZ-10 is the clearest instance in TDR-2025-41 of a control that exists on paper and is weaker in practice.

Reading Occasional-exposure hazard HZ-02 against the verification evidence, the reviewer confirmed mapping at four of six sites and could not evidence the remaining two. On history, register entry HZ-02 has produced no field event in the reviewed period. Robot contacts a pedestrian emerging from a blind cross-aisle is the hazard entered at HZ-02. Occasional-exposure hazard HZ-02 illustrates why TDR-2025-41 separates the severity of a finding from the confidence behind it.

On the register's own scale, register entry HZ-15 is Serious injury severity, Remote exposure, Medium controllability. The design measure recorded at Remote-exposure hazard HZ-15 is egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal. Register entry HZ-15 is the entry for the case that robot obstructs an emergency egress route. Serious injury hazard HZ-15 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The reviewer's note on hazard HZ-16 in the register is that the ring pause defect in July 2025 did not produce a mixed-version robot. Two rollback drills plus a fleet version audit is what verified hazard HZ-16. Register entry HZ-16 produced two near-miss reports from site staff, which is the strongest available test of the control. register entry HZ-16 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Protection at HZ-09 comes from ramp speed limits, a slip detector and a mechanical brake hold. On the register's own scale, hazard HZ-09 in the register is Serious injury severity, Remote exposure, Medium controllability. What the

register does not say about Serious injury hazard HZ-09, and the reviewer does, is that the slip detector is prototyped and unvalidated as at the review date. register entry HZ-09 carries a follow-up request that was satisfied after the draft of TDR-2025-41 was circulated.

Where hazard HZ-12 in the register bites is every robot variant currently in production. HZ-12 is addressed by reverse motion audible warning plus a rear sensing field, and the reviewer traced that control into the design record. One caveat attaches to HZ-12: one site's dock layout leaves less clearance than the layout guide requires. hazard HZ-12 in the register is treated as material to the investment case rather than as a routine engineering item.

Hazard HZ-22 is not clean: the tested duration is shorter than the five-hour outage seen in April 2026. Entry HZ-22 in the hazard register reads: site network loss leaves robots operating without fleet coordination. HZ-22 has never activated in the field and is evidenced only on the bench, and the register was not updated to reflect it. Occasional-exposure hazard HZ-22 was one of two items where TDR-2025-41 could not obtain the underlying data within the engagement.

The record shows that Improbable-exposure hazard HZ-05 has produced no event, though the control has activated repeatedly. Exposure under HZ-05 extends to the manipulator-equipped variant only. Method of verification on hazard HZ-05: witnessed actuation test on every unit at build and at annual service. hazard HZ-05 appears twice in the working papers for TDR-2025-41 because the first answer was corrected after a follow-up request.

Schedule N - Hazard identification and risk assessment register (TDR-2025-41)

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
HZ-01	Robot contacts a pedestrian while travelling in a shared aisle	Serious injury	Frequent	Low	speed and separation monitoring with a protective stop on margin breach	witnessed approach testing on six robots plus a review of the on-robot safety log	Tolerable with the measured approach speed assumption stated
HZ-02	Robot contacts a pedestrian emerging from a blind cross-aisle	Serious injury	Occasional	Low	aisle-end speed reduction plus a mirrored sensing field at mapped blind corners	site-specific commissioning test at each mapped blind corner	Tolerable at commissioned sites; two sites have unmapped corners
HZ-03	Load falls from an elevated deck during transport	Serious injury	Occasional	Medium	payload mass and centre-of-gravity limits enforced by the scale interface	bench drop testing plus a payload limit audit across the fitted fleet	Tolerable where the weighing deck is fitted
HZ-04	Gripper closes on a hand during a manual intervention	Serious injury	Remote	Low	force limiting with a mechanical release and an interlock on the access panel	force bench test after every calibration, evidenced by a calibration record	Tolerable once calibration is gated in the service workflow
HZ-05	Drive power is not removed when the emergency stop is actuated	Fatality	Improbable	Low	hardwired dual-channel stop chain independent of the compute module	witnessed actuation test on every unit at build and at annual service	Broadly acceptable
HZ-06	Robot moves unexpectedly during maintenance	Serious injury	Occasional	Low	maintenance mode with a physical lockout and a two-step exit confirmation	procedure walk-through with three service technicians and a lockout inventory	Tolerable
HZ-07	Robot travels into an unmapped area and contacts fixed plant	Equipment damage	Occasional	Medium	map keep-out enforcement plus a boundary confidence check in localization	simulation coverage of boundary cases plus a site survey audit	Tolerable where survey quality is confirmed

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
HZ-08	Two robots collide at a corridor intersection	Equipment damage	Frequent	High	corridor reservation with deadlock detection and a deterministic yield tie-break	reservation protocol replay plus the pinch-point simulation scenario	Tolerable
HZ-09	Robot fails to stop on a descending ramp when a wheel loses traction	Serious injury	Remote	Medium	ramp speed limits, a slip detector and a mechanical brake hold	ramp testing on a wet and a dry surface at two sites	Not yet tolerable without the slip detector
HZ-10	Charging contact arcs during an unplanned disconnect	Fire	Improbable	Low	contactor sequencing with an arc-suppression design in the charge dock	supplier test report plus an internal witnessed disconnect test	Broadly acceptable
HZ-11	Battery pack thermal runaway during opportunity charging	Fire	Improbable	Low	cell temperature limits, pack-level current derating and a dock thermal cutout	supplier abuse test report plus a review of pack fault records	Broadly acceptable, subject to the state-of-health drift item
HZ-12	Operator is struck by a robot reversing out of a charge dock	Minor injury	Occasional	Low	reverse motion audible warning plus a rear sensing field	witnessed reverse test plus a review of the dock layout at four sites	Tolerable
HZ-13	Robot ignores a temporary barrier placed by site staff	Serious injury	Occasional	Medium	detection of unmapped obstacles with a stop, plus a site procedure for barriers	obstacle detection bench test plus a site procedure review	Tolerable where the barrier meets the visibility guidance
HZ-14	Remote operator issues a motion command to the wrong robot	Minor injury	Occasional	Low	explicit robot selection with a live video confirmation before motion	console interaction test plus a review of session records	Tolerable once the stale roster defect is closed
HZ-15	Robot obstructs an emergency egress route	Serious injury	Remote	Medium	egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal	site commissioning check plus a fire drill observation at two sites	Tolerable
HZ-16	Firmware update leaves a robot in a mixed-version state	Equipment damage	Remote	Low	atomic update with rollback and a version consistency check before mission acceptance	two rollback drills plus a fleet version audit	Tolerable
HZ-17	Safety controller fault log overwrites evidence needed for an investigation	No direct harm	Frequent	Low	periodic fault log extraction to the cloud with a retention policy	log extraction record review across the fleet	Tolerable, with an evidence-loss caveat
HZ-18	Speed reduction engages too late for a fast-approaching person	Serious injury	Occasional	Low	protective distance calculation with a stated approach speed assumption	bench approach testing at the assumed speed and at a higher speed	Not yet tolerable at the higher approach speed
HZ-19	Robot enters a dock occupied	Serious injury	Remote	Low	dock occupancy	occupancy sensor bench	Tolerable

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
	by a person				sensing plus a low approach speed inside the dock envelope	test plus a witnessed dock approach	
HZ-20	Lidar contamination degrades detection without a fault being raised	Serious injury	Occasional	Medium	return-intensity monitoring with a cleaning prompt and a degraded-mode speed limit	contamination bench test plus a review of cleaning compliance at three sites	Tolerable where cleaning compliance is evidenced
HZ-21	Manual push of a powered robot causes an uncontrolled movement	Minor injury	Remote	Low	push detection that engages the brake and requires a deliberate release	witnessed push test on two robot variants	Broadly acceptable
HZ-22	Site network loss leaves robots operating without fleet coordination	Equipment damage	Occasional	Medium	local yield behaviour with a conservative speed limit and a timed stop	network loss injection at one site plus a bench network partition test	Tolerable for outages under the tested duration

3. Risk assessment parameters and residual risk

Method of verification on HZ-18: bench approach testing at the assumed speed and at a higher speed. Occasional-exposure hazard HZ-18 covers the case where speed reduction engages too late for a fast-approaching person. HZ-18 applies to the two pallet-mover variants only. Occasional-exposure hazard HZ-18 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

For hazard HZ-06 the verifying activity was procedure walk-through with three service technicians and a lockout inventory. Register entry HZ-06 identifies the hazard that robot moves unexpectedly during maintenance. Occasional-exposure hazard HZ-06 is addressed by maintenance mode with a physical lockout and a two-step exit confirmation, and the reviewer traced that control into the design record. Occasional-exposure hazard HZ-06 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

On history, register entry HZ-07 produced a single event before the current control was fitted. Against HZ-07 the team relies on map keep-out enforcement plus a boundary confidence check in localization. Robots fitted with the weighing deck is the exposed population for Occasional-exposure hazard HZ-07. HZ-07 is one of the entries where TDR-2025-41 relies on the company's own measurement rather than an independent one.

Verification of register entry HZ-08 was by reservation protocol replay plus the pinch-point simulation scenario. HZ-08 carries an open qualification: the tie-break landed in October 2025 after two stall events. The parameter set for hazard HZ-08 in the register reads Equipment damage, Frequent, High. HZ-08 is reported here in full because a summary of it in an earlier draft of TDR-2025-41 was misread as an assurance.

Post-mitigation, Serious injury hazard HZ-20 sits at Tolerable where cleaning compliance is evidenced and is recorded as partially verified. One qualification on hazard HZ-20 in the register the reviewer would not remove is that cleaning compliance records were incomplete at one site. Hazard HZ-20 scores Serious injury on severity; exposure is Occasional and controllability Medium. hazard HZ-20 would be a candidate for the first post-close engineering review under any reasonable plan.

Where register entry HZ-04 bites is robots fitted with the weighing deck. Hazard HZ-04 was tested by force bench test after every calibration, evidenced by a calibration record, and the reviewer read the resulting record. HZ-04 is mitigated by force limiting with a mechanical release and an interlock on the access panel. Serious injury hazard HZ-04 would be a candidate for the first post-close engineering review under any reasonable plan.

Serious injury hazard HZ-01 is addressed by speed and separation monitoring with a protective stop on margin breach, and the reviewer traced that control into the design record. Evidence of verification at Serious injury hazard HZ-01 consists of witnessed approach testing on six robots plus a review of the on-robot safety log. Register entry HZ-01

produced two near-miss reports from site staff, which is the strongest available test of the control. hazard HZ-01 is the clearest instance in TDR-2025-41 of a control that exists on paper and is weaker in practice.

Reading hazard HZ-11 against the verification evidence, state-of-health drift on opportunity-charged packs is an open modelling item. On history, register entry HZ-11 produced one field event that did not result in harm. Mitigation for register entry HZ-11 is cell temperature limits, pack-level current derating and a dock thermal cutout. hazard HZ-11 in the register has no single accountable owner, and that is the finding the reviewer would press hardest.

4. Safety-related subsystems and their verification

Taking the evidence together, Charge Scheduler the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Charge Scheduler carries 7 open defects, among them a case where opportunity-charge windows are not released when a shift ends earlier than planned. Evidence examined for Charge Scheduler comprised the scheduler policy document and four sites' charge utilisation exports. The interface Charge Scheduler publishes is versioned, and the fleet allocator and the fleet health monitor were checked for use of any undocumented field. Placement of Charge Scheduler is cloud, regional, with a conservative local fallback policy on the robot, which the reviewer tested by tracing a request end to end. Charge Scheduler was reviewed against the position twelve months earlier, and the direction of travel is favourable.

On end-of-life exposure, Charge Scheduler shows none; the dependency set was refreshed in the 2026 platform upgrade. Charge Scheduler pairs a Python 3.11 implementation with cloud, containers on a managed Kubernetes cluster, and its data lives in PostgreSQL 14, shared instance with the allocator. Where Charge Scheduler is concerned, the commitment is that fleet state of charge stays above 35 per cent through the peak shift at every site; the reviewer traced the metric to its source. Coverage for Charge Scheduler was 63 per cent line coverage at the review date, and the trend over the reviewed period was flat, with a one-off drop when a test suite was quarantined. Fleet Services releases Charge Scheduler fortnightly, without a formal approval gate until March 2026, and no release in the sampled window bypassed the gate set. Charge Scheduler was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Service-level intent for Edge Gateway Agent states that no telemetry is lost for an outage shorter than four hours. Edge Gateway Agent was tested against the dependency ageing register, which records none; the transport library was updated in early 2026. The Edge Gateway Agent codebase is Rust 1.79 throughout, deployed onto on-robot Linux, a single supervised process, and its durable record is a local write-ahead buffer sized for four hours. Ownership of Edge Gateway Agent sits with Platform Infrastructure; Sam Oyelaran signs off changes to it. For Edge Gateway Agent the reviewer records that the reviewer accepts the engineering position while noting that the verifying test does not exist. Edge Gateway Agent is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Edge Gateway Agent publishes a stable interface, though two fields used by the telemetry ingest path and the robot shadow store are documented only in code comments. Edge Gateway Agent carries 5 open defects, among them a case where buffer pressure is reported as a robot-level metric and is not aggregated to a site view. The topology for Edge Gateway Agent is edge, one per robot, with mutual authentication to the cloud, and the reviewer confirmed it against the deployment manifests. Platform Infrastructure releases Edge Gateway Agent every three weeks, and no release in the sampled window bypassed the gate set. The dependency map places the on-robot services that emit telemetry above Edge Gateway Agent and the telemetry ingest path and the robot shadow store below it. Edge Gateway Agent carries a follow-up request that was satisfied after the draft of TDR-2025-41 was circulated.

Hosting for Traffic Manager is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to Redis 7 for corridor reservations with a PostgreSQL 14 audit trail. On Traffic Manager the conclusion is qualified: the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. The interface Traffic Manager publishes is versioned, and the motion planner and the remote operations console were checked for use of any undocumented field. The open defect backlog for Traffic Manager holds 12 items, the most consequential being that a corridor reservation leaks when a robot drops off the network mid-traverse. Traffic Manager reports 74 per cent line coverage, measured by the same instrumentation the pipeline uses. Traffic Manager was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The ageing position for Traffic Manager is Redis 6 client library still vendored in one code path, and the reviewer checked it against the vendor statements on file. The fleet allocator and the site map service feed Traffic Manager; a failure in Traffic Manager is felt by the motion planner and the remote operations console. Underlying Traffic Manager in this report are the reservation protocol specification and a replay of three deadlock events. Changes to Traffic Manager reach production weekly, paired with the allocator release. The topology for Traffic Manager is cloud, regional, with a degraded on-robot yield behaviour if the link drops, and the reviewer confirmed it against the deployment manifests. Traffic Manager is the clearest instance in TDR-2025-41 of a control that exists on paper and is weaker in practice.

17 items sit in the Fleet Health Monitor backlog; the entry that bears on the investment case is that alert grouping collapses distinct faults at the same site into one case, hiding the second fault. What the reviewer had for Fleet Health Monitor was the alert rule inventory, three months of case records and a rule coverage review; the conclusions below do not reach past it. Fleet Health Monitor shows 57 per cent line coverage in the pipeline, and coverage alone overstates the assurance here, because the bench suite is the real safety net. Taking the evidence together, Fleet Health Monitor the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. End-of-life review of Fleet Health Monitor found none identified. Fleet Health Monitor was reviewed against the position twelve months earlier, and nothing measurable has changed.

Fleet Health Monitor owns its own data and exposes it only through a versioned interface, which the reviewer read. Changes to Fleet Health Monitor reach production fortnightly. Fleet Health Monitor is owned by the Field Reliability team, with Wanda Njoku named as the engineering owner of record. Hosting for Fleet Health Monitor is cloud, containers on a managed Kubernetes cluster, and its Python 3.11 code writes through to a time-series store plus PostgreSQL 14 for case records. Placement of Fleet Health Monitor is cloud, regional, which the reviewer tested by tracing a request end to end. Fleet Health Monitor was reviewed against the position twelve months earlier, and nothing measurable has changed.

Deployment of Costmap Fusion is edge, co-scheduled with the planner on the compute module, which sets the blast radius of any defect in it. Release cadence for Costmap Fusion is monthly, batched with perception model updates, and the release records bear that out. For Costmap Fusion the reviewer records that the component is adequate today and carries a supportability liability that is not on the funded plan. The open defect backlog for Costmap Fusion holds 21 items, the most consequential being that a stale lidar frame is retained for one extra cycle while a driver reconnects. Ownership of Costmap Fusion sits with Perception; Daniel Okafor signs off changes to it. Costmap Fusion was excluded from an earlier draft of TDR-2025-41 and reinstated after the reviewer obtained the supporting record.

58 per cent line coverage is reported for Costmap Fusion, and the reviewer re-ran the measurement on a clean checkout. Costmap Fusion carries a single measurable objective, namely that fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms. Interface discipline around Costmap Fusion is adequate; the schema is versioned and change is gated on a compatibility check. A shared-memory ring buffer with no durable persistence is the system of record for Costmap Fusion, a C++17 component running on on-robot Linux, pinned to the drive control core. Upstream of Costmap Fusion sit the lidar driver, the depth camera bridge and the wheel odometry filter; downstream, the motion planner and the speed and separation monitor consume its output. Costmap Fusion was raised again with the owning engineer after the initial interview, and the position did not change.

Gripper Controller shows 41 per cent line coverage in the pipeline, and the reviewer regards that as adequate for the risk this component carries. Of the 15 open items against Gripper Controller, the reviewer treats one as material: force limiting relies on a calibration that is only re-run when a gripper is replaced. Evidence examined for Gripper Controller comprised the gripper control specification, calibration records for eleven units and a force bench test. A local parameter store is the system of record for Gripper Controller, a C++17 component running on on-robot Linux with a firmware coprocessor. In production Gripper Controller is edge, only on the two robot variants fitted with a manipulator, unchanged since the last architecture review. Gripper Controller is reported here in full because a summary of it in an earlier draft of TDR-2025-41 was misread as an assurance.

Gripper Controller was examined for unsupported components; none; the coprocessor part is current production. The order execution path and the safety controller fault input rely on Gripper Controller, which in turn relies on the payload scale interface and the perception grasp proposals. Gripper Controller is promoted quarterly, with the ring order fixed by policy. Gripper Controller publishes a stable interface, though two fields used by the order execution path and the safety controller fault input are documented only in code comments. Ines Duarte of Robot Firmware maintains

Gripper Controller, and the service inventory entry was current at the review date. Gripper Controller is reported here in full because a summary of it in an earlier draft of TDR-2025-41 was misread as an assurance.

Payload Scale Interface depends on the load cell hardware and is depended on by the gripper controller and the order execution path. Payload Scale Interface publishes a stable interface, though two fields used by the gripper controller and the order execution path are documented only in code comments. What the reviewer had for Payload Scale Interface was the interface specification, calibration records for nine units and a module-swap test; the conclusions below do not reach past it. Deployment of Payload Scale Interface is edge, only on robots fitted with the weighing deck, which sets the blast radius of any defect in it. Against Payload Scale Interface the backlog runs to 11 items, and it includes a known behaviour whereby the calibration record is stored locally and is lost when the compute module is swapped. Payload Scale Interface was checked against the company's own register, which had not been updated since the change landed.

Test coverage on Payload Scale Interface is 36 per cent line coverage and the measurement basis was confirmed with Ines Duarte. Changes to Payload Scale Interface reach production as needed. For Payload Scale Interface the supportability position is none identified. The objective recorded against Payload Scale Interface is that payload mass is reported within 2 per cent of the reference load, and it is reported weekly to the engineering leadership. The reviewer's judgement on Payload Scale Interface is that the component would not block the transaction, and it would appear on any competent post-close engineering plan. Payload Scale Interface was reviewed against the position twelve months earlier, and nothing measurable has changed.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-41)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Motion Planner	Rust 1.79	Autonomy Core	71%	plan cycle completes inside 40 ms at the 99th percentile	the costmap fusion pipeline and the localization engine	none; the toolchain is pinned to a supported compiler release
Costmap Fusion	C++17	Perception	58%	fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms	the lidar driver, the depth camera bridge and the wheel odometry filter	a vendored geometry library whose upstream branch left support in 2024
Fleet Allocator	Go 1.22	Fleet Services	82%	task assignment inside 900 ms of order arrival at p95	the order broker and the robot shadow store	none identified in the reviewed manifests
Traffic Manager	Go 1.22	Fleet Services	74%	corridor deadlock detected and broken inside two seconds	the fleet allocator and the site map service	Redis 6 client library still vendored in one code path
Charge Scheduler	Python 3.11	Fleet Services	63%	fleet state of charge stays above 35 per cent through the peak shift at every site	the battery management service and the site shift calendar	none; the dependency set was refreshed in the 2026 platform upgrade
Safety Controller	C, written to a constrained coding subset	Safety Engineering	91%	a validated protective-stop trigger de-energises the drive inside 120 ms	the E-stop chain, the speed and separation monitor and the drive current sensors	none; the supplier's support commitment runs to 2031
E-Stop Chain	VHDL	Safety Engineering	witnessed test	drive power is removed within 250 ms of button actuation on every tested unit	the operator stop buttons and the perimeter interlocks	none; the relay parts are current production
Lidar Driver	C++17	Robot Firmware	44%	no more than one dropped scan per ten thousand at nominal spin rate	the sensor hardware and the time synchronisation daemon	the vendor's software development kit is two major versions behind current
Localization Engine	C++17	Autonomy Core	66%	pose error stays under 25 mm in	the lidar driver, the wheel	an unmaintained scan-matching

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
				mapped aisles at survey speed	odometry filter and the map distribution channel	library is still linked in the fallback path
WMS Connector	Java 21	Integrations	69%	order acknowledgement returned to the warehouse system inside 1.5 s at p95	the customer warehouse management systems and the identity service	one customer integration still runs on a connector build from 2024
Order Broker	Go 1.22	Fleet Services	78%	no accepted order is lost, measured by journal reconciliation each night	the WMS connector and the site configuration service	none identified
Telemetry Ingest	Go 1.22	Data and Telemetry	61%	telemetry visible in the operator console inside 30 s of emission at p95	the edge gateway agent and the event bus	the stream processor runs a release that leaves community support in late 2026
OTA Updater	Rust 1.79 on the robot, Go 1.22 in the cloud	Platform Infrastructure	76%	an update is either fully applied or fully rolled back, with no mixed-version robot	the release orchestrator and the artefact signing service	none; artefact signing was moved to the current key format in 2025
Identity and Authorization	Go 1.22	Security Engineering	84%	token issuance inside 200 ms at p95 and no unauthenticated write path	the customer directory federations and the secret store	none identified in the reviewed manifests
Remote Operations Console	TypeScript	Applications	52%	operator input reaches the robot inside 350 ms at p95 on a supported network	the identity service, the telemetry ingest path and the video snippet service	a charting library that no longer receives security updates
Simulation Harness	Python 3.11	Developer Productivity	47%	the nightly scenario suite completes inside four hours	the planner and localization builds under test	the physics core depends on a library branch that stopped receiving fixes in 2023
Hardware-in-the-Loop Rig	Python 3.11 test harness around production firmware	Robot Firmware	39%	every firmware candidate runs the full bench suite before ring one	the firmware candidate artefacts and the bench sensor stimulators	the rig controller runs an operating system release that left vendor support in 2025

4. Safety-related subsystems and their verification - part 2 of 2

Below, the 4. Safety-related subsystems and their verification continues; nothing separates this part from the last except the page on which it happens to fall. Nothing is restated from the earlier part.

Definitions given once for the 4. Safety-related subsystems and their verification are relied on again rather than restated, the compilation basis being identical. The break is editorial.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-41) (continued, part 2 of 2)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Fleet Health Monitor	Python 3.11	Field Reliability	57%	a fleet-affecting condition raises an alert inside three minutes of first symptom	the telemetry ingest path and the robot shadow store	none identified
Battery Management Service	C on the pack controller, Python 3.11 in the cloud	Robot Firmware	54%	pack cell imbalance is reported before it reaches the derating threshold	the pack sensors and the charge scheduler	one pack revision uses a cell supplier that has since discontinued the part

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Gripper Controller	C++17	Robot Firmware	41%	a grasp completes or reports failure inside 1.2 s	the payload scale interface and the perception grasp proposals	none; the coprocessor part is current production
Map Authoring Studio	TypeScript	Applications	49%	a published map revision reaches every robot at the site inside fifteen minutes	the site survey imports and the identity service	a geometry rendering library one major version behind its supported line
Site Provisioning Service	Go 1.22	Platform Infrastructure	64%	a new site reaches a working configuration inside one working day	the identity service and the customer network intake records	none identified
Event Bus	operated open-source broker, no in-house code	Platform Infrastructure	witnessed test	publish acknowledgement inside 25 ms at p95 and no partition without a leader	every publishing service	the broker minor release in production leaves the vendor's support window in 2027
Config Distribution	Rust 1.79 agent	Platform Infrastructure	72%	a configuration change reaches every online robot inside five minutes	the site provisioning service and the artefact signing service	none; the bundle format was revised in 2025
Robot Shadow Store	Go 1.22	Fleet Services	68%	shadow state is no more than two seconds behind the robot at p95	the telemetry ingest path and the edge gateway agent	none identified
Analytics Warehouse Loader	Python 3.11	Data and Telemetry	45%	the daily load completes before 06:00 local time at every region	the telemetry ingest path, the order broker and the billing usage meter	one transformation still runs on a deprecated warehouse function
Billing Usage Meter	Go 1.22	Data and Telemetry	79%	metered usage reconciles to the order journal within 0.5 per cent each month	the order broker and the WMS connector	none identified
Alarm Router	Python 3.11	Field Reliability	51%	a dispatched alarm reaches an on-call responder inside 90 s at p95	the fleet health monitor and the on-call rotation source	the paging provider's older application interface is still in use
Video Snippet Service	Go 1.22	Applications	43%	a requested snippet is available to the operator inside 45 s	the on-robot camera buffer and the identity service	a transcoder build with known but low-severity upstream advisories
Speed and Separation Monitor	C++17	Safety Engineering	87%	commanded speed is reduced before the separation distance falls below the protective distance	the costmap fusion pipeline and the safety-rated scanner inputs	none; the sensing hardware is current production
Wheel Odometry Filter	C++17	Robot Firmware	62%	odometry drift stays under 1.5 per cent of distance travelled on a clean floor	the wheel encoders and the inertial measurement unit	none identified
Payload Scale Interface	C++17	Robot Firmware	36%	payload mass is reported within 2 per cent of the reference load	the load cell hardware	none identified
Edge Gateway Agent	Rust 1.79	Platform Infrastructure	74%	no telemetry is lost for an outage shorter than four	the on-robot services that emit telemetry	none; the transport library was updated in

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
				hours		early 2026
Release Orchestrator	Go 1.22	Developer Productivity	55%	no artefact reaches production without a recorded approval and a passing gate set	the build system, the simulation harness gate and the identity service	none identified

5. Safety-relevant incidents and protective-stop behaviour

Kestrel Medical Supply incident INC-2025-110 traces back to a single fact: the loader treated a missing partition as empty rather than as incomplete. What Analytics Warehouse Loader incident INC-2025-110 tells the reviewer is that management's severity rating is defensible and its remediation priority is not. Incident INC-2025-110 was detected because the customer challenged the report three days later. Time to mitigate data integrity incident INC-2025-110 was not applicable; reporting only, and time to resolve was 18 days. The corrective action for INC-2025-110 is to fail the load on an incomplete partition and reprocess the affected days, and 2025-10-31 is the date the team gave. incident INC-2025-110 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

On data integrity event INC-2025-110 the evidence is moderate on the cause and strong on the delivery position, which was checked in the repository; the severity assessment is stated separately below. Kestrel Medical Supply incident INC-2025-110 affected 1 site or tenant with no robot taken out of service. In taxonomy terms data integrity incident INC-2025-110 is a data integrity incident in Analytics Warehouse Loader. The reviewer checked the INC-2025-110 commitment against the code and release records; landed on 2025-09-21 with a reprocessing run for the affected week. Contributing to incident INC-2025-110, and not repaired by the remediation, is that the partition completeness check was written but never wired into the load gate. Kestrel Medical Supply incident INC-2025-110 was excluded from an earlier draft of TDR-2025-41 and reinstated after the reviewer obtained the supporting record.

In availability event INC-2025-071, the upgrade runbook did not require a partition leadership check between broker restarts, which is a process finding rather than a code finding. The affected population in INC-2025-071 was 6 sites and 214 robots. The reviewer grades the availability event INC-2025-071 evidence as moderate on the cause and strong on the delivery position, which was checked in the repository and has not upgraded the finding to compensate. The remediation committed after Telemetry Ingest incident INC-2025-071 was to add a leadership and under-replicated partition check to the broker upgrade runbook, owned by Erik Palmgren with a due date of 2025-07-31. Attention came to availability incident INC-2025-071 when the ingest availability alert fired within ninety seconds. availability incident INC-2025-071 rests on a single source, and the reviewer has said so rather than implying corroboration.

Availability incident INC-2025-071 is recorded as follows on delivery: landed on 2025-07-18 and exercised in the September failover drill. Underneath availability event INC-2025-071 lies the design decision that a broker partition lost its leader during a rolling upgrade of the streaming cluster. 2025-06-11 is the date of multiple sites incident INC-2025-071, on which a regional ingest endpoint failed over and edge buffers filled during a fifty-two minute outage. Drawing the evidence in multiple sites incident INC-2025-071 together, the team's analysis was honest about a gap it could have concealed. INC-2025-071 belongs to the availability group and sits with Telemetry Ingest. availability incident INC-2025-071 is the clearest instance in TDR-2025-41 of engineering practice being better than the documentation suggests.

2025-04-21 is the date of scheduling event INC-2025-052, on which an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. Landed on 2025-06-11 with a nightly reconciliation report is the position on the incident INC-2025-052 commitment as at the review date. The affected population in scheduling event INC-2025-052 was one site and 44 robots. 4 h 10 m elapsed before INC-2025-052 was mitigated, with resolution at 16 days. After scheduling event INC-2025-052, Nadia Semenova took the action to release charge windows on a shift-end event and reconcile the calendar nightly by 2025-06-30. incident INC-2025-052 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

INC-2025-052 classifies as a scheduling event against Charge Scheduler. Underneath Evergreen Warehousing incident INC-2025-052 lies the design decision that the scheduler treated a shift calendar entry as immutable once the

window had opened. Strong, since the reviewer reproduced the failure mode in a laboratory environment is the evidence grade for scheduling incident INC-2025-052, and the reader should not read severity into it. In incident INC-2025-052 the first signal was the site's own shift report, escalated the next morning, and the reviewer treats that as reported detection. Evergreen Warehousing incident INC-2025-052 carries a second-order finding: the site had changed its shift pattern without notifying the deployment team. Charge Scheduler incident INC-2025-052 is the reason one section of TDR-2025-41 concludes narrowly rather than generally.

INC-2025-124 was raised on 2025-10-01 after a shard rebalance served shadow state up to ninety seconds stale without marking it stale. Consistency incident INC-2025-124 reached the engineering team when the console latency alert fired, though the underlying cause took a day to attribute. The post-mortem for incident INC-2025-124 identifies the cause as follows: the rebalance path bypassed the staleness annotation applied on the normal read path. The commitment register shows INC-2025-124 against an undertaking to annotate staleness on every read path and add a rebalance-under-load test to the platform suite, with Marta Kellerman named and 2025-11-30 committed. The affected population in Robot Shadow Store incident INC-2025-124 was 5 sites and 178 robots. consistency incident INC-2025-124 was raised again with the owning engineer after the initial interview, and the position did not change.

The contributing factor recorded against consistency event INC-2025-124 is that no test exercised a shard rebalance while the cluster was under load. Verification of the multiple sites incident INC-2025-124 remediation gives landed on 2025-10-30; the load test was added on 2025-11-12. Grading incident INC-2025-124 on evidence rather than on consequence gives limited, being a single account corroborated only by the ticket timeline. Consistency incident INC-2025-124 classifies as a consistency event against Robot Shadow Store. The judgement recorded against INC-2025-124 is that the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. consistency event INC-2025-124 has no single accountable owner, and that is the finding the reviewer would press hardest.

Beyond the direct cause, observability event INC-2026-028 shows that case closure did not check for unresolved constituent alerts. Granite Peak Distribution incident INC-2026-028 affected one site and 22 robots. Time to mitigate INC-2026-028 was not a service outage; 2 days of delayed repair, and time to resolve was 31 days. Drawing the evidence in observability incident INC-2026-028 together, the customer-reported detection path is the finding, not the defect itself. For incident INC-2026-028 the follow-through was tested rather than assumed, and landed on 2026-04-10. observability event INC-2026-028 appears twice in the working papers for TDR-2025-41 because the first answer was corrected after a follow-up request.

Grading Fleet Health Monitor incident INC-2026-028 on evidence rather than on consequence gives moderate; the narrative is consistent but the reviewer could not independently reproduce it. Incident INC-2026-028 classifies as a observability event against Fleet Health Monitor. 2026-03-10 is the date of Granite Peak Distribution incident INC-2026-028, on which two distinct faults at one site were grouped into a single case and the second went unworked for two days. Underneath incident INC-2026-028 lies the design decision that alert grouping keyed on site and severity rather than on the underlying signal. How observability incident INC-2026-028 surfaced matters to the assessment: the customer asked why one symptom persisted after the case had been closed. Fleet Health Monitor incident INC-2026-028 illustrates why TDR-2025-41 separates the severity of a finding from the confidence behind it.

Evergreen Warehousing incident INC-2026-045 supports one conclusion and not another: the incident is unremarkable in itself and revealing about the observability of the surrounding system. Duration for incident INC-2026-045 splits into 5 h 12 m of impact and 26 days to close the underlying cause. For observability incident INC-2026-045 the reviewer holds limited, being a single account corroborated only by the ticket timeline evidence and reports the finding at that strength. On delivery, Evergreen Warehousing incident INC-2026-045 shows this: the alert landed on 2026-05-17; the buffer extension is in progress. Classifying observability event INC-2026-045 as observability matters because the Edge Gateway Agent backlog contains two further items of the same kind. observability incident INC-2026-045 is the reason one section of TDR-2025-41 concludes narrowly rather than generally.

After incident INC-2026-045, Sam Oyelaran took the action to aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current hardware by 2026-06-30. Root cause in observability event INC-2026-045 was that the buffer is sized for four hours and drops oldest-first without raising an alert. The reviewer treats one contributing factor in Edge Gateway Agent incident INC-2026-045 as the more important lesson: buffer pressure was never aggregated to a site view, so the condition was not observable. Evergreen Warehousing felt Edge Gateway Agent incident

INC-2026-045, which affected one site and 39 robots. On 2026-04-21 the record for observability event INC-2026-045 opens: a site-wide network change left robots buffering for five hours and the oldest telemetry samples were discarded. INC-2026-045 would be a candidate for the first post-close engineering review under any reasonable plan.

Duration for tooling event INC-2026-021 splits into not applicable; developer tooling of impact and open at the review date to close the underlying cause. Incident INC-2026-021 would have been shorter or smaller had it not been that the build pinned the compiler but not the optimisation flags used for release builds. The corrective action for tooling event INC-2026-021 is to pin the full build configuration and add a determinism check to the harness, and 2026-05-31 is the date the team gave. Detection came through the autonomy team noticed a step change in the pass rate, which places Simulation Harness incident INC-2026-021 on the reported side of the ledger. Simulation Harness owns tooling incident INC-2026-021, which the reviewer classes as tooling. internal incident INC-2026-021 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

INC-2026-021 began on 2026-02-24 when a nightly simulation suite produced non-reproducible results after the physics core was rebuilt. Internal incident INC-2026-021 supports one conclusion and not another: an operational workaround masked the defect for longer than the incident record suggests. For incident INC-2026-021 the reviewer holds moderate; the narrative is consistent but the reviewer could not independently reproduce it evidence and reports the finding at that strength. Root cause in incident INC-2026-021 was that the physics core is not deterministic across optimisation levels. Simulation Harness incident INC-2026-021 was contained to no production site and no robot directly, the exposure being to internal systems. Simulation Harness incident INC-2026-021 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Detection came through the payload damage was reported by the customer, which places Blue Mesa Components incident INC-2026-009 on the reported side of the ledger. Gripper Controller incident INC-2026-009 is not read as an isolated failure: the elapsed resolution time reflects a scheduling decision rather than an engineering difficulty. Blue Mesa Components felt incident INC-2026-009, which affected one site and 4 robots. Time to mitigate calibration incident INC-2026-009 was 1 h 30 m to quarantine the affected robots, and time to resolve was 48 days. Gripper Controller incident INC-2026-009 produced one commitment: to require a calibration record before returning a manipulator robot to service, owned by Ines Duarte, dated 2026-04-15. Blue Mesa Components incident INC-2026-009 was cross-checked against the engineering record supplied for TDR-2025-41, and the two agree.

For Gripper Controller incident INC-2026-009 the follow-through was tested rather than assumed, and landed on 2026-03-16 with a workflow gate. A contributing factor in incident INC-2026-009 was that the service workflow allowed the step to be marked complete without a calibration record. Calibration incident INC-2026-009 records that on 2026-01-27, a gripper applied more force than the configured limit after a mechanical replacement without recalibration. Gripper Controller incident INC-2026-009 was caused by a design gap: force limiting depends on a calibration that is only triggered by a service workflow step. For incident INC-2026-009 the reviewer holds moderate, resting on the team's own post-mortem and one supporting log extract evidence and reports the finding at that strength. Blue Mesa Components incident INC-2026-009 has no single accountable owner, and that is the finding the reviewer would press hardest.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2025-41)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-200 tote mover	network radio	2.18.6	14	both rigs	current
HR-410 manipulator	E-stop harness	4.18.3	8	rig two only	one revision behind
HR-300 pallet mover	charge contact assembly	4.3.8	7	rig one only	one revision behind
HR-240 tote mover	load cell deck	3.11.8	3	neither rig	current
HR-410 manipulator	network radio	4.2.0	13	neither rig	superseded
HR-320 pallet mover	network radio	3.17.1	10	rig two only	superseded
HR-240 tote mover	wheel encoder set	1.13.5	11	rig one only	superseded
HR-200 tote mover	compute module	3.17.5	13	both rigs	pre-production
HR-300 pallet mover	depth camera	4.10.4	5	neither rig	current
HR-200 tote mover	charge contact assembly	3.14.8	4	neither rig	pre-production
HR-240 tote mover	network radio	1.9.2	4	rig two only	superseded
HR-410 manipulator	inertial measurement unit	1.17.2	2	both rigs	superseded
HR-320 pallet mover	gripper coprocessor	1.12.2	6	rig one only	one revision behind
HR-320 pallet mover	load cell deck	4.7.5	6	rig two only	one revision behind

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-200 tote mover	wheel encoder set	4.10.7	11	neither rig	superseded
HR-320 pallet mover	load cell deck	2.1.5	13	rig one only	superseded
HR-410 manipulator	depth camera	3.2.5	4	neither rig	current
HR-410 manipulator	depth camera	1.5.7	5	neither rig	one revision behind
HR-200 tote mover	gripper coprocessor	4.5.2	4	rig two only	superseded
HR-300 pallet mover	charge contact assembly	4.14.3	10	neither rig	pre-production
HR-300 pallet mover	wheel encoder set	3.4.0	6	rig one only	current
HR-200 tote mover	gripper coprocessor	2.16.3	6	neither rig	current

5. Safety-relevant incidents and protective-stop behaviour - part 2 of 2

Continuing the 5. Safety-relevant incidents and protective-stop behaviour, the rows below follow directly on from part 2 minus one, with no gap in the reference sequence. The sequence is unbroken either side of it.

Treat the conventions declared for the 5. Safety-relevant incidents and protective-stop behaviour as governing here too; this part introduces none of its own. The sequence is unbroken either side of it.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2025-41) (continued, part 2 of 2)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-300 pallet mover	depth camera	4.15.4	4	both rigs	one revision behind
HR-320 pallet mover	inertial measurement unit	1.1.7	6	neither rig	superseded
HR-240 tote mover	navigation lidar	1.18.1	10	rig one only	current
HR-240 tote mover	gripper coprocessor	3.9.0	10	rig two only	pre-production
HR-320 pallet mover	wheel encoder set	1.0.1	5	neither rig	current
HR-410 manipulator	load cell deck	3.17.1	13	both rigs	one revision behind
HR-240 tote mover	charge contact assembly	3.4.4	7	rig one only	one revision behind
HR-410 manipulator	gripper coprocessor	4.11.6	2	both rigs	pre-production
HR-320 pallet mover	wheel encoder set	2.2.8	3	both rigs	one revision behind
HR-240 tote mover	depth camera	1.13.2	12	rig two only	superseded
HR-410 manipulator	network radio	4.14.2	13	rig two only	superseded
HR-320 pallet mover	safety scanner	4.15.7	5	rig one only	pre-production
HR-320 pallet mover	safety scanner	1.10.2	14	neither rig	superseded
HR-300 pallet mover	depth camera	1.6.3	9	neither rig	superseded
HR-410 manipulator	safety scanner	4.1.0	7	rig two only	superseded
HR-410 manipulator	E-stop harness	1.13.2	9	rig two only	one revision behind
HR-200 tote mover	inertial measurement unit	3.7.8	4	rig two only	superseded
HR-200 tote mover	network radio	2.4.0	4	rig two only	one revision behind
HR-300 pallet mover	load cell deck	2.13.6	9	rig one only	current
HR-300 pallet mover	wheel encoder set	1.4.4	9	rig two only	pre-production
HR-410 manipulator	network radio	3.2.5	8	both rigs	superseded
HR-320 pallet mover	E-stop harness	4.0.2	14	both rigs	current

6. Verification method and evidence behind each control

Severity at hazard HZ-03 is Serious injury, with the exposure rated Occasional and controllability Medium. After mitigation, hazard HZ-03 in the register is Tolerable where the weighing deck is fitted; the register status is partially verified. The scope of hazard HZ-03 is every robot variant currently in production. hazard HZ-03 illustrates why TDR-2025-41 separates the severity of a finding from the confidence behind it.

Residual risk at hazard HZ-17 in the register is stated as Tolerable, with an evidence-loss caveat, and the status is open. The verification method for hazard HZ-17 is log extraction record review across the fleet. Where No direct harm hazard HZ-17 bites is the newest three sites, where commissioning is incomplete. Frequent-exposure hazard HZ-17 is one of the entries where TDR-2025-41 relies on the company's own measurement rather than an independent one.

On history, HZ-14 produced two near-miss reports from site staff. What the register does not say about HZ-14, and the reviewer does, is that the roster invalidation fix landed in July 2025. Where register entry HZ-14 bites is the sites that have moved to opportunity charging. Occasional-exposure hazard HZ-14 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Minor injury hazard HZ-21 was scoped to every robot variant currently in production, and the reviewer checked the scoping against the fleet record. Hazard HZ-21 in the register is addressed by push detection that engages the brake and requires a deliberate release, and the reviewer traced that control into the design record. After mitigation, register entry HZ-21 is Broadly acceptable; the register status is verified. HZ-21 appears twice in the working papers for TDR-2025-41 because the first answer was corrected after a follow-up request.

The verification method for Remote-exposure hazard HZ-19 is occupancy sensor bench test plus a witnessed dock approach. Hazard HZ-19 in the register is the entry for the case that robot enters a dock occupied by a person. Serious injury hazard HZ-19 was scoped to the two pallet-mover variants only, and the reviewer checked the scoping against the fleet record. Remote-exposure hazard HZ-19 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

On the register's own scale, HZ-13 is Serious injury severity, Occasional exposure, Medium controllability. Reading register entry HZ-13 against the verification evidence, two sites use barriers below the guidance height and detection is then marginal. Hazard HZ-13 depends for its mitigation on detection of unmapped obstacles with a stop, plus a site procedure for barriers. hazard HZ-13 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2025-41)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
give every corridor reservation a lease with an expiry and add a site-level queue depth alert	INC-2025-041	Owen Brantley	2025-05-30	landed	no evidence located
re-derive the time offset on every reconnect and add the affected variant to the second bench ri	INC-2025-047	Yuki Harada	2025-06-15	partly landed	interview only
release charge windows on a shift-end event and reconcile the calendar nightly	INC-2025-052	Nadia Semenova	2025-06-30	landed	test added and observed
key suppression on the tenant and identifier pair and backfill the affected order records	INC-2025-058	Tomas Villareal	2025-07-15	landed	interview only
make the allocator consult charger availability before excluding a robot	INC-2025-063	Marta Kellerman	2025-07-31	landed	commit inspected
add a leadership and under-replicated partition check to the broker upgrade runbook	INC-2025-071	Erik Palmgren	2025-07-31	landed	commit inspected
invalidate the roster cache on a fleet membership event and add a session-level regression test	INC-2025-078	Anjali Sekhon	2025-08-31	landed	runbook change inspected
persist the pause state and require a design note for any change to the release state machine	INC-2025-084	Erik Palmgren	2025-08-15	landed	commit inspected
add the connector torque check to incoming inspection and retrofit the affected harness batch	INC-2025-090	Bettina Vogel	2025-09-30	partly landed	no evidence located
replace the mailbox fallback with a	INC-2025-096	Wanda Njoku	2025-10-15	partly landed	release record inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
secondary responder and import rotations from the scheduling					
add a deterministic tie-break and return the pinch-point scenario to the nightly suite	INC-2025-103	Priya Raghunathan	2025-10-31	partly landed	test added and observed
fail the load on an incomplete partition and reprocess the affected days	INC-2025-110	Farida Cisse	2025-10-31	landed	configuration export inspected
redact authorization headers at the logging layer and cut build log retention to fourteen days	INC-2025-118	Sonia Krall	2025-10-31	partly landed	commit inspected
annotate staleness on every read path and add a rebalance-under-load test to the platform suite	INC-2025-124	Marta Kellerman	2025-11-30	landed	runbook change inspected
restrict export to published revisions and shorten the publication path so the workaround is unn	INC-2025-131	Lucia Ferrante	2025-12-15	partly landed	test added and observed
split control traffic onto a dedicated topic with its own quota and revisit the topic inventory	INC-2025-139	Erik Palmgren	2026-01-31	partly landed	interview only
fold the reset into the re-run command and make double-counting detectable in the ledger	INC-2025-146	Nadia Semenova	2025-12-31	landed	interview only
apply the lifecycle rule through the configuration baseline and delete the over-retained clips	INC-2025-152	Anjali Sekhon	2026-01-31	partly landed	commit inspected
replace the fallback matcher and re-survey the affected aisles	INC-2026-004	Rahul Menon	2026-06-30	not landed	configuration export inspected
require a calibration record before returning a manipulator robot to service	INC-2026-009	Ines Duarte	2026-04-15	landed	commit inspected
move precondition validation to apply time and add an offline apply case to the platform suite	INC-2026-015	Sam Oyelaran	2026-03-31	landed	test added and observed
pin the full build configuration and add a determinism check to the harness	INC-2026-021	Colin Achebe	2026-05-31	not landed	release record inspected
group on the signal and block closure while a constituent alert remains active	INC-2026-028	Wanda Njoku	2026-04-30	landed	no evidence located
measure closing rate where the sensing permits and re-run the protective distance calculation	INC-2026-033	Bettina Vogel	2026-08-31	not landed	test added and observed

Commitment	Source	Owner	Committed date	Status	Evidence inspected
serialise cancellation against assignment and add an hourly orphan sweep	INC-2026-039	Grace Lindqvist	2026-05-31	landed	runbook change inspected
aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current h	INC-2026-045	Sam Oyelaran	2026-06-30	partly landed	interview only
check every template against the baseline and re-converge the four forked templates	INC-2026-051	Marcus Duplessis	2026-06-15	partly landed	interview only
refit the estimator on opportunity-charge data and add an imbalance alert independent of state o	INC-2026-057	Peter Halvorsen	2026-09-30	not landed	runbook change inspected
require a justification and a second approver within one working day for every emergency release	INC-2026-062	Colin Achebe	2026-08-31	not landed	runbook change inspected
add a slip detector that flags the condition and issue a site advisory on cleaning windows	INC-2026-068	Yuki Harada	2026-09-15	not landed	runbook change inspected
Partition the fleet allocator write path	TD-01	engineering leadership	2027-03-31	in design	register entry only
Replace the unmaintained scan-matching library in the localization fallback	TD-02	engineering leadership	2026-12-31	committed	register entry only
Retire the vendored geometry library in costmap fusion	TD-03	engineering leadership	2027-06-30	backlog	register entry only
Raise robot software line coverage above sixty per cent on the perception path	TD-04	engineering leadership	2027-03-31	in progress	register entry only
Bring the second bench rig to the current sensor configuration	TD-05	engineering leadership	2026-09-30	committed	register entry only
Make the nightly simulation suite blocking	TD-06	engineering leadership	2026-11-30	blocked	register entry only

7. Safety-related engineering work not yet funded

Progress on support item TD-09 is recorded as committed against a 2027-06-30 date. The estimate for TD-09 is 1.5 engineer-quarters and $196,000. The reviewer's qualification on programme item TD-09 is that the upgrade requires a maintenance window at every site. register item TD-09 is the clearest instance in TDR-2025-41 of a control that exists on paper and is weaker in practice.

Behind a platform change that has its own target date is where safety item TD-19 sits in the programme. The most significant open safety item in the reviewed record is the point the reviewer would put to management on TD-19. Whether Speed and Separation Monitor item TD-19 happens depends on funding, and it is unfunded and dependent on capital rather than engineering effort. safety item TD-19 is included because a reader of TDR-2025-41 should be able to disagree with the inference without repeating the work.

The safety entry at programme item TD-20 is to validate and ship the wheel slip detector. 2026-09-15 is the target for register item TD-20, whose status is in progress. Left in place, the condition behind Wheel Odometry Filter item

TD-20 drift on wet floors is neither corrected nor flagged. register item TD-20 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

TD-06 on Simulation Harness carries this observation: blocked on the physics core determinism work. Quality item TD-06 is not funded, and carried in the register regardless; the register status and the funding position are not the same thing. The estimate for programme item TD-06 is 1 engineer-quarter and $115,000. TD-06 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-41.

Safety evidence item TD-18 is funded only if the team's headcount plan is met in the engineering plan the reviewer was shown. Register item TD-18 carries this observation: the safety team supports it; it is not funded in the current plan. The safety evidence entry at TD-18 on Safety Controller is to aggregate the safety controller fault log to the cloud continuously. programme item TD-18 is the clearest instance in TDR-2025-41 of engineering practice being better than the documentation suggests.

TD-08 on Remote Operations Console is sequenced last in a three-item chain, so its date is the least reliable on the register. Progress on security supportability item TD-08 is recorded as backlog against a 2027-01-31 date. The business consequence recorded against programme item TD-08 is that it no upstream security updates for a customer-facing component. TD-08 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The configuration entry at Site Provisioning Service item TD-11 is to re-converge the four forked site provisioning templates. The estimate for register item TD-11 is 0.5 engineer-quarters and $61,000. TD-11 appears in the register and is not funded, and carried in the register regardless. programme item TD-11 was excluded from an earlier draft of TDR-2025-41 and reinstated after the reviewer obtained the supporting record.

A small item that keeps returning as a field support cost is the point the reviewer would put to management on programme item TD-22. 0.25 engineer-quarters is the estimate on quality item TD-22, which converts to $42,000. Reading the programme plan, Payload Scale Interface item TD-22 is ahead of two other items that depend on it. TD-22 on Payload Scale Interface is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The business consequence recorded against IP hygiene item TD-24 is that it three modified files under a file-level copyleft licence appear unpublished. On funding, register item TD-24 is funded in the current financial year. Costmap Fusion item TD-24 is sequenced behind a customer maintenance window that is not yet agreed. programme item TD-24 is treated as material to the investment case rather than as a routine engineering item.

If TD-14 is not done, the exposure is that it bench queueing is the largest term in firmware lead time. Requires capital for a third rig rather than engineering effort is the point the reviewer would put to management on Hardware-in-the-Loop Rig item TD-14. The register entry programme item TD-14 reads: introduce a second bench rig queue to cut firmware lead time, in the delivery category. delivery item TD-14 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Progress on Fleet Allocator item TD-01 is recorded as in design against a 2027-03-31 date. Reading the programme plan, register item TD-01 is last in a three-item chain, so its date is the least reliable on the register. Scalability item TD-01 is not straightforward: the load-test series identifies this as the single binding scalability constraint. TD-01 on Fleet Allocator illustrates why TDR-2025-41 separates the severity of a finding from the confidence behind it.

On funding, register item TD-17 is unfunded and dependent on capital rather than engineering effort. Progress on security item TD-17 is recorded as in design against a 2026-11-30 date. Item register item TD-17 in the debt register is to enforce a two-person rule on OTA artefact signing, and it bears on OTA Updater. TD-17 on OTA Updater is not material on its own and belongs in this report because of what it reveals about the surrounding process.

TD-03 is partly funded, with the second phase unbudgeted in the engineering plan the reviewer was shown. TD-03 is not straightforward: management ranked this behind the separation monitoring work. Backlog is the status of programme item TD-03, targeted at 2027-06-30. TD-03 on Costmap Fusion is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

8. Safety-rated components and third-party assessment

For otel-instrument the obligation is to hold a current commercial licence for each developer seat, and the reviewer tested whether it is met. Because otel-instrument is used tracing across cloud services, its licence terms reach the delivered software. Under Apache-2.0, otel-instrument is treated as no copyleft in the inventory. otel-instrument is the clearest instance in TDR-2025-41 of a control that exists on paper and is weaker in practice.

Wms-adapter-sdk version 2.2.0 is catalogued at OSS-017 and is used one customer's warehouse system adapter. Under commercial, per-seat, wms-adapter-sdk is treated as no copyleft in the inventory. Wms-adapter-sdk shows an approval status of approved 2023-12-05 against the company's own policy. wms-adapter-sdk would move from a concern to a verified defect if one further measurement were supplied.

OSS-028 records k8s-operator-framework 1.31.0 against Site Provisioning Service, cloud deployment controllers. In addition to its terms, k8s-operator-framework brings a fork that must be maintained locally for as long as it is used. K8s-operator-framework reaches the customer only through Site Provisioning Service, and the reviewer confirmed there is no second path. k8s-operator-framework carries a follow-up request that was satisfied after the draft of TDR-2025-41 was circulated.

Weak copyleft applies to geom2d-fusion, whose licence is LGPL-2.1. The approval position on geom2d-fusion is approved 2023-08-04. Costmap Fusion is where geom2d-fusion is consumed, which determines whether distribution occurs at all. geom2d-fusion was cross-checked against the engineering record supplied for TDR-2025-41, and the two agree.

Provenance of rtos-shim is public registry, signature verified. What the company must actually do about rtos-shim is to attribute the component in the customer-facing notice file. Rtos-shim presents an upstream project that is active but changes interface frequently to the engineering team. rtos-shim was checked against the corresponding entry in the company's own register, and the register was current.

Protobuf-runtime entered the codebase by this route: public registry, signature verified. Protobuf-runtime is licensed under BSD-3-Clause, with copyleft exposure recorded as no. Beyond licensing, protobuf-runtime carries a fork that must be maintained locally for as long as it is used. protobuf-runtime illustrates why TDR-2025-41 separates the severity of a finding from the confidence behind it.

Meeting the terms on safety-controller-toolchain means nothing beyond retaining the licence text with the source. The terms applying to safety-controller-toolchain are commercial, development use only; the copyleft classification is no. OSS-019 records safety-controller-toolchain 3.0.4 against Safety Controller, safety controller build toolchain. safety-controller-toolchain is treated as material to the investment case rather than as a routine engineering item.

Tracing pager-provider-sdk to its source gives supplied by the paging provider. Support risk on pager-provider-sdk takes the form of an upstream project with no active maintainer. Pager-provider-sdk carries a practical obligation: to keep the dynamic-linking boundary intact. pager-provider-sdk was tested rather than accepted, and the test is described in the working papers behind TDR-2025-41.

Support risk on cell-model-lib takes the form of a commercial supplier whose support statement expires inside the plan period. The licence on cell-model-lib is Apache-2.0; copyleft exposure is no. Battery Management Service is where cell-model-lib is consumed, which determines whether distribution occurs at all. cell-model-lib rests on a single source, and the reviewer has said so rather than implying corroboration.

Version 0.3.1 of wheel-slip-model is the catalogued build at OSS-025, used prototype slip detector, not shipped. Wheel-slip-model carries a practical obligation: to keep the dynamic-linking boundary intact. Wheel Odometry Filter is where wheel-slip-model is consumed, which determines whether distribution occurs at all. wheel-slip-model was inspected in the repository rather than described from the ticket, which is why the account is specific.

What the company must actually do about kafka-client is to publish the modified files it distributes. Beyond licensing, kafka-client carries a build-time dependency whose absence would stop the pipeline rather than the product. The open question on kafka-client is that one service still vendors a 6.x client. kafka-client has no single accountable owner, and that is the finding the reviewer would press hardest.

9. Security findings with a safety consequence

The reason Low-severity finding SEC-2026-10 was not downgraded is that an export could be altered after the fact. Remediated is the status of Low-severity finding SEC-2026-10, dated 2026-06-05. For Low-severity finding SEC-2026-10 the record shows that bucket configuration read from the cloud export, which is moderate, because the demonstration was performed in a laboratory tenant rather than in production evidence and is graded separately from severity. Billing Usage Meter finding SEC-2026-10 arrives as a Low finding with a 4.0 score, and the reviewer did not adjust either. Low-severity finding SEC-2026-10 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Residual risk after remediation of SEC-2025-12 is medium; management accepted the risk with a compensating audit trail. Reading Low-severity finding SEC-2025-12, the defect is that analytics warehouse service account held write access to raw telemetry; the discovery date is 2025-09-25. The practical consequence in Analytics Warehouse Loader finding SEC-2025-12 is that a loader defect or compromise could alter source telemetry. SEC-2025-12 stands accepted; not planned is what the record gives for closure. SEC-2025-12 is reported at the evidence strength available and has not been promoted to fit the narrative.

On 2025-09-17 the tester recorded finding SEC-2025-02: build logs printed the full authorization header. Assessment finding SEC-2025-02 is rated High with a numeric band of 8.1 on the tester's scoring scale. In finding SEC-2025-02 the exploitation path means any engineer with build log access could replay a service credential. Substantiating High-severity finding SEC-2025-02 is the fact that the tester extracted a working token from a retained log, graded strong. assessment finding SEC-2025-02 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Motion Planner finding SEC-2026-17 is not fully retired by its remediation: low; the port requires a signed local credential. High-severity finding SEC-2026-17 translates into the following exposure: a local compromise could alter planner limits without detection. Evidence behind SEC-2026-17 against Motion Planner is that override applied on a laboratory robot and observed in plan output, which the reviewer treats as moderate, because the demonstration was performed in a laboratory tenant rather than in production. On 2026-04-22 the tester recorded Motion Planner finding SEC-2026-17: motion planner parameter override accepted from an unauthenticated local port. Motion Planner finding SEC-2026-17 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The finding at Medium-severity finding SEC-2025-03, dated 2025-09-18, is that remote operations session relay accepted an unpinned certificate chain. On finding SEC-2025-03 the tester's evidence was that demonstrated in the tester's laboratory network, not on a customer site, and the reviewer grades it moderate. Remote Operations Console finding SEC-2025-03 arrives as a Medium finding with a 6.2 score, and the reviewer did not adjust either. Assessment finding SEC-2025-03 translates into the following exposure: an on-path attacker on a customer network could observe operator commands. assessment finding SEC-2025-03 was one of two items where TDR-2025-41 could not obtain the underlying data within the engagement.

Rating for Order Broker finding SEC-2026-13 is Low, band 3.3; the reviewer re-read the scoring rationale and accepted it. The finding at finding SEC-2026-13, dated 2026-04-20, is that order broker accepted an oversized payload without a limit. SEC-2026-13 against Order Broker produces this judgement: the severity is right and the ownership is unclear, since two teams each believe the other holds it. Remediation status on SEC-2026-13 against Order Broker is remediated; the record shows 2026-05-15. SEC-2026-13 against Order Broker was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Low-severity finding SEC-2026-11 was graded Low, band 4.4, by Meridian Assurance engagement MA-2026-1. Low-severity finding SEC-2026-11 rests on the observation that file permissions observed on a laboratory robot; the reviewer holds that to be strong. The reviewer declines to soften SEC-2026-11 against Gripper Controller: the underlying pattern appears in two other components that the tester did not reach. Committed 2026-11-30 is the closure position on Low-severity finding SEC-2026-11, which the ledger records as open. Gripper Controller finding SEC-2026-11 was one of two items where TDR-2025-41 could not obtain the underlying data within the engagement.

Residual risk after remediation of assessment finding SEC-2026-08 is low; role grants now require a second approver and are alerted. The reason assessment finding SEC-2026-08 was not downgraded is that one administrator can

grant themselves tenant-wide access. The finding at Identity and Authorization finding SEC-2026-08, dated 2026-04-18, is that identity service admin role assignable without a second approval. Assessment finding SEC-2026-08 arrives as a Medium finding with a 6.4 score, and the reviewer did not adjust either. Medium-severity finding SEC-2026-08 rests on a single source, and the reviewer has said so rather than implying corroboration.

The reviewer declines to soften finding SEC-2026-07: the item is open past its committed date and no revised date has been agreed. The residual position on Medium-severity finding SEC-2026-07 is medium; network segmentation is the only current control, stated after mitigation rather than assuming full success. As at the review date SEC-2026-07 against Hardware-in-the-Loop Rig was open and the closure entry reads committed 2026-10-31. Hardware-in-the-Loop Rig finding SEC-2026-07 translates into the following exposure: an unpatched laboratory host sits on the engineering network. Hardware-in-the-Loop Rig finding SEC-2026-07 is reported here in full because a summary of it in an earlier draft of TDR-2025-41 was misread as an assurance.

At 5.9 the score on assessment finding SEC-2025-07 places it in the Medium group. Finding SEC-2025-07 records that site provisioning stored a shared bootstrap secret in the template, discovered on 2025-09-22. In finding SEC-2025-07 the exploitation path means the same secret was reusable at any site provisioned from that template. After the fix, assessment finding SEC-2025-07 carries residual risk described as medium; two forked templates were not re-issued until May 2026. Medium-severity finding SEC-2025-07 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Partially remediated is the status of OTA Updater finding SEC-2025-09, dated pending. Rating for finding SEC-2025-09 is Medium, band 6.3; the reviewer re-read the scoring rationale and accepted it. The finding at SEC-2025-09 against OTA Updater, dated 2025-09-24, is that oTA artefact signing keys held by three engineers without a quorum control. A single key holder could sign an unreviewed firmware artefact is the exposure OTA Updater finding SEC-2025-09 describes. Medium-severity finding SEC-2025-09 has no single accountable owner, and that is the finding the reviewer would press hardest.

10. Open safety qualifications carried forward

10 matters remain open at the date of TDR-2025-41, and none of them is closed by the passage of time alone. K. Vandermeer has dated each open item in the present report so that a reader can see which have been open longest rather than only how many there are. The interview programme behind TDR-2025-41 covered 9 teams and produced 9 follow-up requests, of which most were satisfied. The reviewer found the interview evidence for this report candid, and in two cases more candid than the written record. K. Vandermeer interviewed 15 people for the report at TDR-2025-41; where an account was uncorroborated it is marked as such.

The present report keeps declined items visible: risk acceptance without an expiry date is itself recorded as a finding. K. Vandermeer asked in every declined case who would revisit the decision and when, and TDR-2025-41 records the answer even where there was none. The approach K. Vandermeer took in report TDR-2025-41 was to test the company's own assertions rather than to restate them. Access for report TDR-2025-41 was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. The report at TDR-2025-41 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

K. Vandermeer has left several concerns in report TDR-2025-41 at limited evidence strength rather than dropping them or dressing them up. The present report reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. The open items in the present report are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. The reviewer has resisted the temptation to tidy the present report: the open items are open because the evidence is not there yet. The functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet produced more evidence than the report at TDR-2025-41 reproduces, and the working papers hold the remainder.

5 indicators are recommended in TDR-2025-41; the reviewer deliberately avoided proposing anything that needs a new pipeline. Report TDR-2025-41 would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. Evidence gathering for the present report was iterative: an initial document request, 15 interviews, then a second request that closed most of the gaps. TDR-2025-41 follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a

verified defect. K. Vandermeer has kept the working papers behind this report in a form that a second reviewer could re-perform.

11. Conclusions on the safety position

Stated without qualification for once, the conclusion of report TDR-2025-41 is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer's conclusion in the present report is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The instruction behind the report at TDR-2025-41 was narrow: examine the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet and report by 2026-02-04. Ardent Peak commissioned report TDR-2025-41 from Calder Rowe Engineering Advisory to test the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet ahead of the Series B financing. K. Vandermeer has resisted rounding the findings towards a comfortable conclusion in either direction.

K. Vandermeer would put 7 items from the present report into the first hundred days rather than into a warranty. The transaction reading of report TDR-2025-41 is that the engineering risk is priced in the plan rather than in the purchase agreement. Monitoring recommended by TDR-2025-41 relies on metrics the company already emits, so no new instrumentation is required to start. Post-close monitoring under this report would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. K. Vandermeer has separated what the company has built from what the company can demonstrate about what it has built.

The monitoring pack this report proposes would have surfaced two of the incidents examined here before a customer noticed. K. Vandermeer would add one qualitative item to the pack proposed in this report: the count of commitments closed without verifying evidence. Where the report at TDR-2025-41 reports a supplier's test result, the reviewer read the certificate and did not witness the test. The reviewer notes that the present report was prepared while three remediation programmes were in flight, so several positions will have moved by completion. K. Vandermeer has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

For Identity and Authorization the stated target is that token issuance inside 200 ms at p95 and no unauthenticated write path, expressed against production traffic rather than a benchmark. The Security Engineering team carries Identity and Authorization on its service inventory, and Sonia Krall answered for it in interview. Data ownership is clear for Identity and Authorization: it writes PostgreSQL 14 with a hardware-backed key store and no other service writes there. Documents and data inspected for Identity and Authorization were the authorization model document, a token lifetime audit and the secret rotation log. Test coverage on Identity and Authorization is 84 per cent line coverage and the measurement basis was confirmed with Sonia Krall. Identity and Authorization appears twice in the working papers for TDR-2025-41 because the first answer was corrected after a follow-up request.

One coupling concern applies to Alarm Router: PostgreSQL 14 for routes and delivery receipts is shared with a second service, so a schema change is not a local decision. Alarm Router releases fortnightly. The evidence base for Alarm Router was the routing table export, delivery receipts for one quarter and two missed-page reviews. Reading the dependency map from Alarm Router outward gives the fleet health monitor and the on-call rotation source on the input side and the paging provider and the customer notification channel on the output side. Placement of Alarm Router is cloud, regional, with a second delivery provider, which the reviewer tested by tracing a request end to end. Alarm Router was one of two items where TDR-2025-41 could not obtain the underlying data within the engagement.